

OS Practical 10

```
nishanth@LAPTOP-6Q3ON07H:~$ mkdir q10
nishanth@LAPTOP-6Q3ON07H:~$ cd q10
nishanth@LAPTOP-6Q3ON07H:~/q10$ echo "Hello inode" > file1.txt
nishanth@LAPTOP-6Q3ON07H:~/q10$ ls -li file1.txt
40262 file1.txt
nishanth@LAPTOP-6Q3ON07H:~/q10$ stat file1.txt
  File: file1.txt
  size: 12          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 40262         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/nishanth)   Gid: ( 1000/nishanth)
Access: 2026-10-01 09:12:31.527736235 +0000
Modify: 2026-10-01 09:12:31.527736235 +0000
Change: 2026-10-01 09:12:31.527736235 +0000
 Birth: 2026-10-01 09:12:31.527736235 +0000
nishanth@LAPTOP-6Q3ON07H:~/q10$ find . -inum 40262
./file1.txt
nishanth@LAPTOP-6Q3ON07H:~/q10$ ln file1.txt hardlink.txt
nishanth@LAPTOP-6Q3ON07H:~/q10$ ls -li file1.txt hardlink.txt
40262 -rw-r--r--  2 nishanth nishanth 12 Oct  1 09:12 file1.txt
40262 -rw-r--r--  2 nishanth nishanth 12 Oct  1 09:12 hardlink.txt
nishanth@LAPTOP-6Q3ON07H:~/q10$ stat file1.txt
  File: file1.txt
  size: 12          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 40262         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/nishanth)   Gid: ( 1000/nishanth)
Access: 2026-10-01 09:12:31.527736235 +0000
Modify: 2026-10-01 09:12:31.527736235 +0000
Change: 2026-10-01 09:13:30.159736605 +0000
 Birth: 2026-10-01 09:12:31.527736235 +0000
nishanth@LAPTOP-6Q3ON07H:~/q10$ echo "Added through hard link" >> hardlink.txt
nishanth@LAPTOP-6Q3ON07H:~/q10$ cat file1.txt
Hello inode
Added through hard link
nishanth@LAPTOP-6Q3ON07H:~/q10$
```

```

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <string.h>

int main() {
    int fd;
    char *data;

    const char *text = "Hello from mmap!\n";
    size_t size = strlen(text);

    fd = open("mmapfile.txt",
              O_RDWR | O_CREAT | O_TRUNC,
              0644);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    if (ftruncate(fd, size) == -1) {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    data = mmap(NULL,
                size,
                PROT_READ | PROT_WRITE,
                MAP_SHARED,
                fd,
                0);

    if (data == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    memcpy(data, text, size);

    printf("Mapped content: %.*s",
           (int)size, data);
}

```

```
msync(data, size, MS_SYNC);

munmap(data, size);
close(fd);

return 0;
}
```

```
nishanth@LAPTOP-6Q3ON07H:~/q10$ nano mmap_demo.c
nishanth@LAPTOP-6Q3ON07H:~/q10$ gcc mmap_demo.c -o mmap_demo
nishanth@LAPTOP-6Q3ON07H:~/q10$ ./mmap_demo
Mapped content: Hello from mmap!
nishanth@LAPTOP-6Q3ON07H:~/q10$ cat mmapfile.txt
Hello from mmap!
nishanth@LAPTOP-6Q3ON07H:~/q10$
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    int fd;
    char buffer[100];

    fd = open("traditional.txt",
              O_RDWR | O_CREAT | O_TRUNC,
              0644);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    const char *text = "Hello from read/write!\n";

    write(fd, text, strlen(text));

    lseek(fd, 0, SEEK_SET);

    ssize_t bytes =
        read(fd, buffer, sizeof(buffer) - 1);

    if (bytes == -1) {
        perror("read");
        close(fd);
        return 1;
    }

    buffer[bytes] = '\0';

    printf("File content: %s", buffer);

    close(fd);

    return 0;
}
```

```
Hello from mmap!  
nishanth@LAPTOP-6Q3ON07H:~/q10$ nano traditional_io.c  
nishanth@LAPTOP-6Q3ON07H:~/q10$ gcc traditional_io.c -o traditional_io  
nishanth@LAPTOP-6Q3ON07H:~/q10$ ./traditional_io  
File content: Hello from read/write!  
nishanth@LAPTOP-6Q3ON07H:~/q10$ time ./mmap_demo  
Mapped content: Hello from mmap!  
  
real    0m0.007s  
user    0m0.000s  
sys     0m0.003s  
nishanth@LAPTOP-6Q3ON07H:~/q10$ time ./traditional_io  
File content: Hello from read/write!  
  
real    0m0.003s  
user    0m0.000s  
sys     0m0.001s  
nishanth@LAPTOP-6Q3ON07H:~/q10$
```